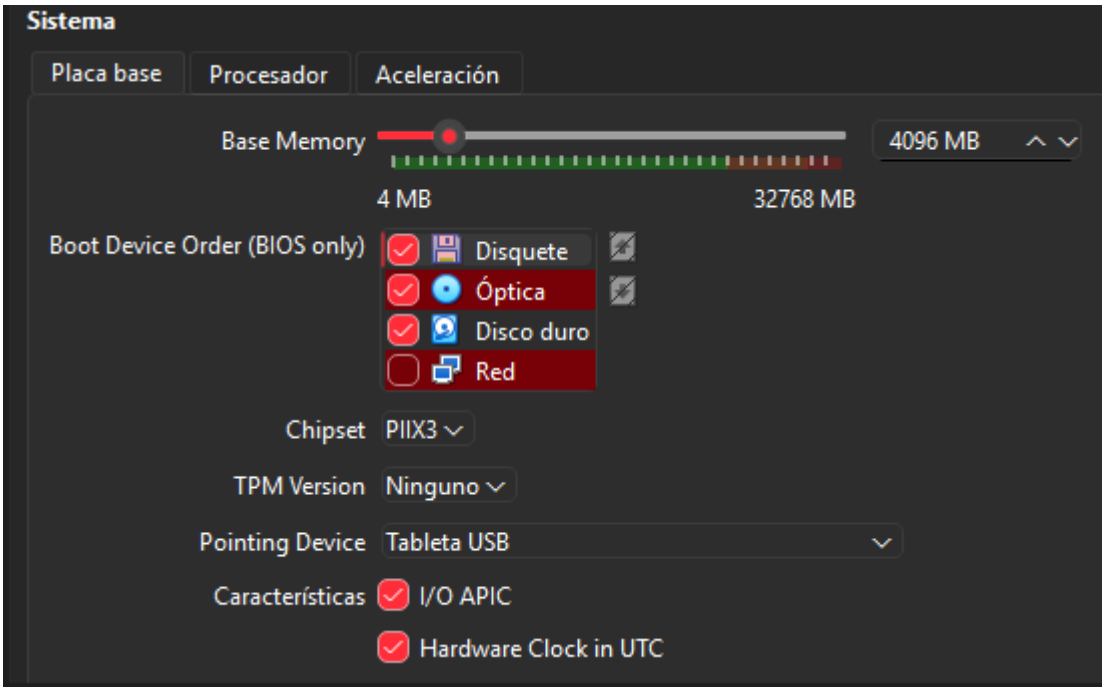
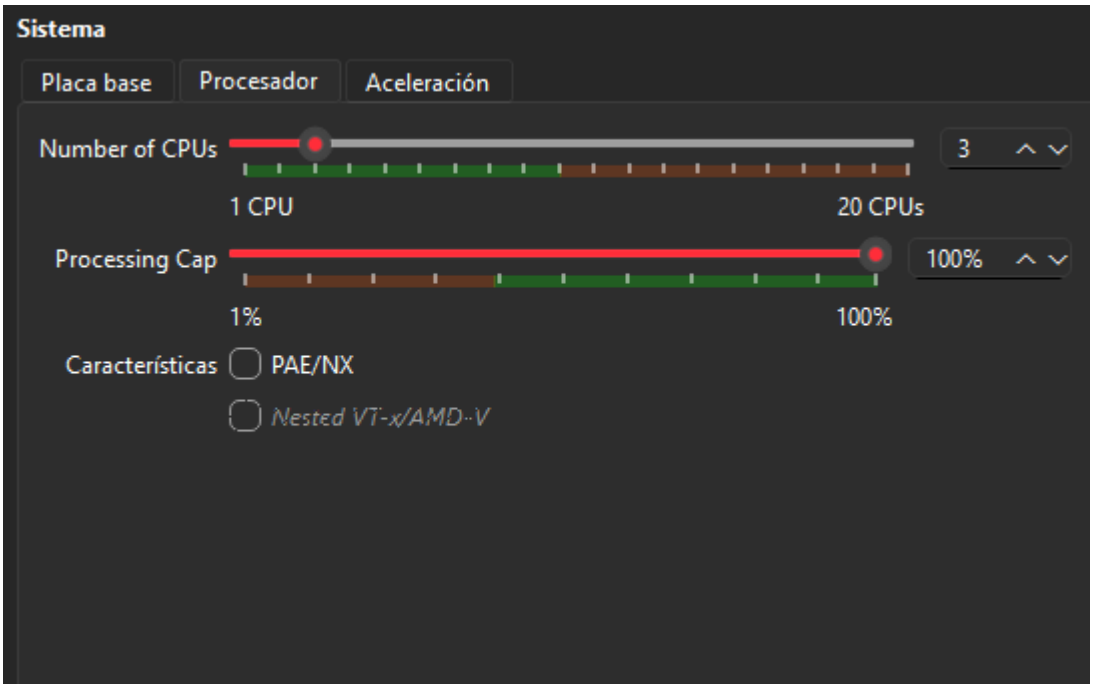
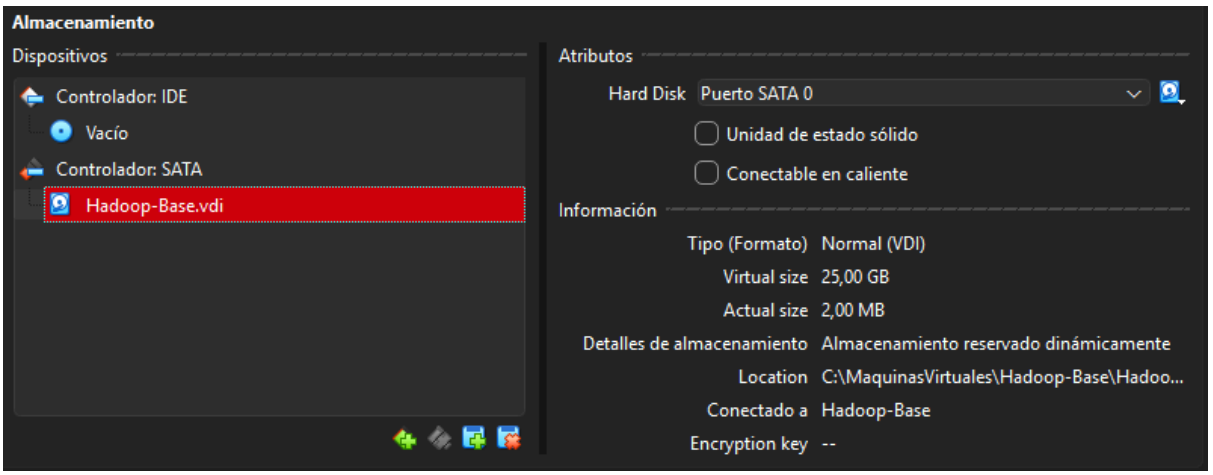
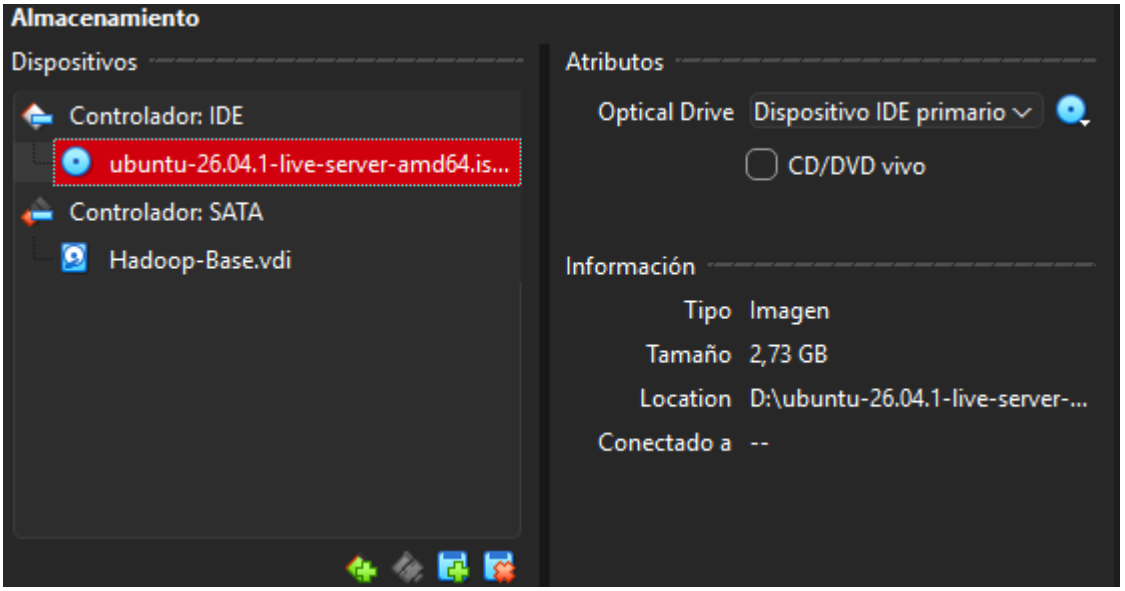
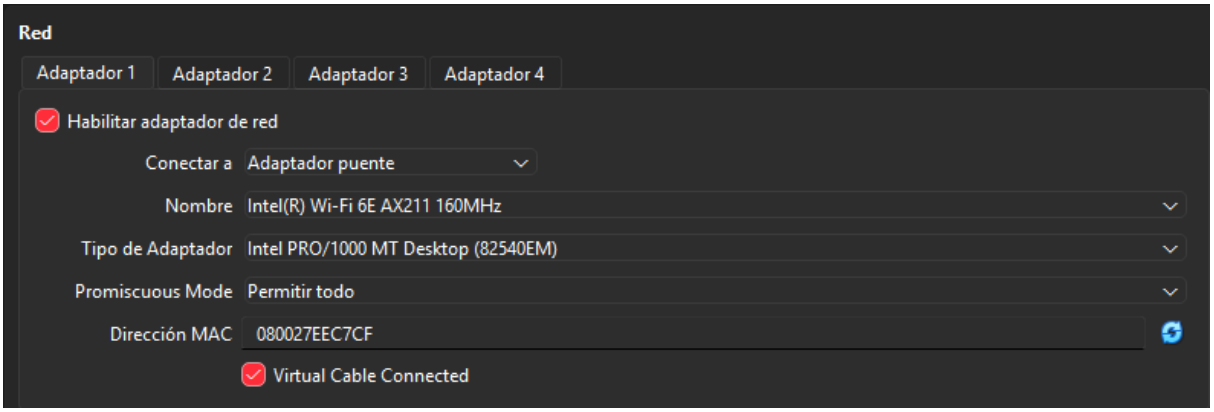
1. Instalación



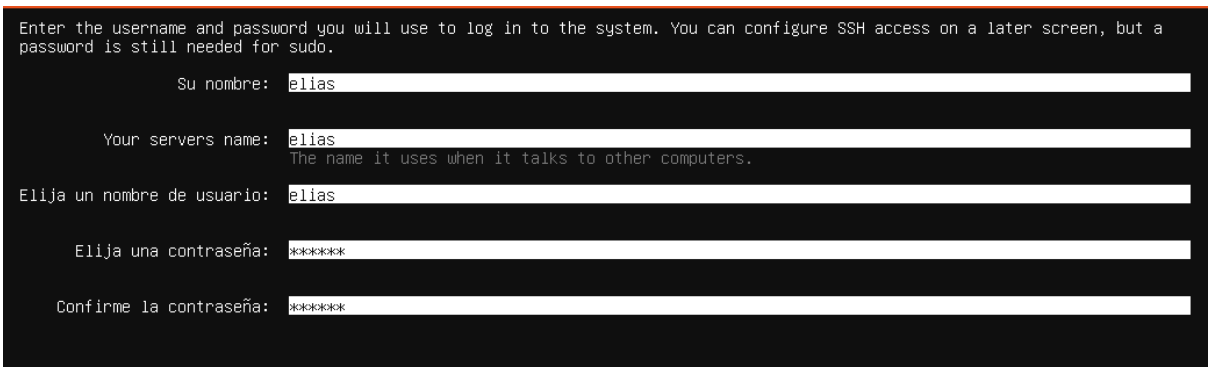



Almacenamiento
Dispositivos
Controlador: IDE
Vacío
Controlador: SATA
Hadoop-Base.vdi
Atributos
Hard Disk
Puerto SATA 0
Unidad de estado sólido
Conectable en caliente
Información
Tipo (Formato) Normal (VDI)
Virtual size 25,00 GB
Actual size 2,00 MB
Detalles de almacenamiento Almacenamiento reservado dinámicamente
Location C:\MaquinasVirtuales\Hadoop-Base\Hadoo...
Conectado a Hadoop-Base
Encryption key --


Almacenamiento
Dispositivos
Controlador: IDE
ubuntu-26.04.1-live-server-amd64.is...
Controlador: SATA
Hadoop-Base.vdi
Atributos
Optical Drive
Dispositivo IDE primario
CD/DVD vivo
Información
Tipo Imagen
Tamaño 2,73 GB
Location D:\ubuntu-26.04.1-live-server-...
Conectado a --


Red
Adaptador 1
Adaptador 2
Adaptador 3
Adaptador 4
Habilitar adaptador de red
Conectar a Adaptador puente
Nombre Intel(R) Wi-Fi 6E AX211 160MHz
Tipo de Adaptador Intel PRO/1000 MT Desktop (82540EM)
Promiscuous Mode Permitir todo
Dirección MAC 080027EEC7CF
Virtual Cable Connected

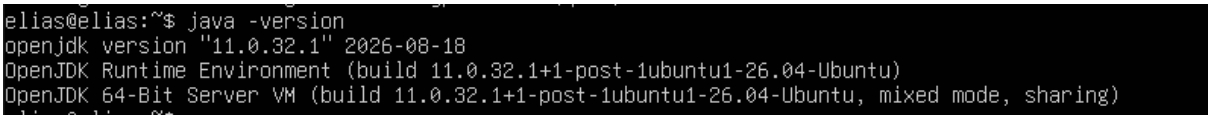
2. Usuarios



contraseña: ab1234

3. Instalación de java y hadoop
   Instalamos java con: sudo apt install openjdk-11-jdk-headless





Instalamos hadoop con: sudo apt install wget openssh-server rsync



Ahora con readlink -f /usr/bin/java muestra la ubicación real del ejecutable de Java. La opción -f sigue los enlaces hasta encontrar el archivo original. Usaremos esa ruta para configurar JAVA_HOME en Hadoop.

```
elias@elias:~$ readlink -f /usr/bin/java
/usr/lib/jvm/java-11-openjdk-amd64/bin/java
```

Ahora con nano -/ .bashrc añadimos al final: export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64

```
# enable color support of ls and also add handy aliases
if [ -x /usr/bin/dircolors ]; then
    test -r ~/.dircolors && eval "$(dircolors -b ~/.dircolors)" || eval "$(dircolors -b
    alias ls='ls --color=auto'
    #alias dir='dir --color=auto'
    #alias vdir='vdir --color=auto'

    alias grep='grep --color=auto'
    alias fgrep='fgrep --color=auto'
    alias egrep='egrep --color=auto'
fi

# colored GCC warnings and errors
#export GCC_COLORS='error=01;31:warning=01;35:note=01;36:caret=01;32:locus=01:quote=01'

# some more ls aliases
alias ll='ls -alF'
alias la='ls -A'
alias l='ls -CF'

# Add an "alert" alias for long running commands.  Use like so:
#   sleep 10; alert
alias alert='notify-send --urgency=low -i "$([ $? = 0 ] && echo terminal || echo error)

# Alias definitions.
# You may want to put all your additions into a separate file like
# ~/.bash_aliases, instead of adding them here directly.
# See /usr/share/doc/bash-doc/examples in the bash-doc package.

if [ -f ~/.bash_aliases ]; then
    . ~/.bash_aliases
fi

# enable programmable completion features (you don't need to enable
# this, if it's already enabled in /etc/bash.bashrc and /etc/profile
# sources /etc/bash.bashrc).
if ! shopt -oq posix; then
  if [ -f /usr/share/bash-completion/bash_completion ]; then
    . /usr/share/bash-completion/bash_completion
  elif [ -f /etc/bash_completion ]; then
    . /etc/bash_completion
  fi
fi
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
```

Y con source ~/.bashrc cargamos los cambios (podemos sustituir el simbolo ~ poniendo “$”)

```
elias@elias:~$ source "$HOME/.bashrc"
elias@elias:~$ _
```

Muestra el valor de JAVA_HOME para comprobar que la ruta quedó configurada.

```
elias@elias:~$ echo "$JAVA_HOME"
/usr/lib/jvm/java-11-openjdk-amd64
```

Para instalar hadoop tenemos que estár en la carpeta HOME
cd “$HOME”

Instalamos hadoop con: wget -c
https://downloads.apache.org/hadoop/common/hadoop-3.4.3/hadoop-3.4.3.tar.gz

```
elias@elias:~$ wget -c https://downloads.apache.org/hadoop/common/hadoop-3.4.3/hadoop-3.4.3.tar.gz
--2026-11-03 10:15:10--  https://downloads.apache.org/hadoop/common/hadoop-3.4.3/hadoop-3.4.3.tar.gz
Resolving downloads.apache.org (downloads.apache.org)... 95.216.224.44, 88.99.208.237, 2a01:4f9:2b:1cc2::2, ...
Connecting to downloads.apache.org (downloads.apache.org)|95.216.224.44|:443... connected.
HTTP request sent, awaiting response... 206 Partial Content
Length: 514917037 (491M), 512573167 (489M) remaining [application/x-gzip]
Saving to: 'hadoop-3.4.3.tar.gz'

hadoop-3.4.3.tar.gz                 100%[===================================================================================>] 491,06M  28,3MB/s    in 19s

2026-11-03 10:15:29 (25,7 MB/s) - ‘hadoop-3.4.3.tar.gz’ saved [514917037/514917037]
```

Vamos a comprobar que el archivo se puede leer:
tar -tzf hadoop-3.4.3.tar.gz > /dev/null

```
elias@elias:~$ tar -tzf hadoop-3.4.3.tar.gz > /dev/null
elias@elias:~$ _
```

Descomprimimos: tar -xzf hadoop-3.4.3.tar.gz

```
elias@elias:~$ tar -xzf hadoop-3.4.3.tar.gz
elias@elias:~$ ls
hadoop-3.4.3  hadoop-3.4.3.tar.gz
```

```
elias@elias:~$ ./hadoop-3.4.3/bin/hadoop version
Hadoop 3.4.3
Source code repository https://github.com/apache/hadoop.git -r 9d50c6884666e794e45102260a4017bb31802e1b
Compiled by stevel on 2026-02-13T14:23Z
Compiled on platform linux-x86_64
Compiled with protoc 3.23.4
From source with checksum 2331238c4c2929e66645316a32a8613
This command was run using /home/elias/hadoop-3.4.3/share/hadoop/common/hadoop-common-3.4.3.jar
elias@elias:~$
```

Abrimos la configuración del usuario: nano "$HOME/.bashrc"
y al final añadimos estas dos líneas:
export HADOOP_HOME=/home/elias/hadoop-3.4.3
export PATH="$PATH:HADOOP_HOME/bin:$HADOOP_HOME/sbin"
La primera guarda la ubicación de Hadoop en `HADOOP_HOME`.
La segunda añade sus carpetas de comandos a `PATH`, para poder ejecutar sus herramientas sin escribir la ruta completa.

```
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
export HADOOP_HOME=/home/elias/hadoop-3.4.3
export PATH="$PATH:$HADOOP_HOME/bin:$HADOOP_HOME/sbin"
```

Volvemos a ejecutar: source "$HOME/.bashrc" para cargar los cambios
mostramos la versión:

```
elias@elias:~$ hadoop version
Hadoop 3.4.3
Source code repository https://github.com/apache/hadoop.git -r 9d50c6884666e794e45102260a4017bb31802e1b
Compiled by stevel on 2026-02-13T14:23Z
Compiled on platform linux-x86_64
Compiled with protoc 3.23.4
From source with checksum 2331238c4c2929e66645316a32a8613
This command was run using /home/elias/hadoop-3.4.3/share/hadoop/common/hadoop-common-3.4.3.jar
```

Ahora con el comando : nano "$HADOOP_HOME/etc/hadoop/hadoop-env.sh" vamos a indicar la ruta de java
y añadiremos al final: export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64

```
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64_
```

Ahora configuraremos dónde estará el NameNode.
nano "$HADOOP_HOME/etc/hadoop/core-site.xml"
y añadimos esto

```
<configuration>
 <property>
        <name>fs.defaultFS</name>
        <value>hdfs://master:9000</value>
 </property>
</configuration>
```

(A partir de aquí lo hice por ssh para poder copiar y pegar)

Ahora configuraremos la replicación y las carpetas de almacenamiento de HDFS
nano "$HADOOP_HOME/etc/hadoop/hdfs-site.xml"

```
<configuration>
  <property>
    <name>dfs.replication</name>
    <value>2</value>
  </property>

  <property>
    <name>dfs.namenode.name.dir</name>
    <value>file:///home/elias/hadoop-data/namenode</value>
  </property>

  <property>
    <name>dfs.datanode.data.dir</name>
    <value>file:///home/elias/hadoop-data/datanode</value>
  </property>
</configuration>
```

Creamos las carpetas que HDFS necesitará para guardar está información:
mkdir -p "$HOME/hadoop-data/namenode""HOME/hadoop-data/datanode"

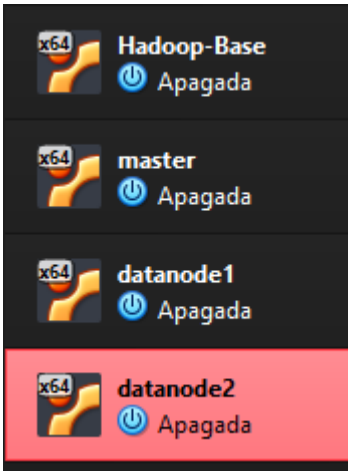
```
elias@elias:~$ mkdir -p "$HOME/hadoop-data/namenode" "$HOME/hadoop-data/datanode"
elias@elias:~$ ls
hadoop-3.4.3  hadoop-3.4.3.tar.gz  hadoop-data
elias@elias:~$ cd hadoop-data
elias@elias:~/hadoop-data$ ls
datanode  namenode
elias@elias:~/hadoop-data$
```

Ahora vamos a preparar la lista de máquinas que ejecutarán los DataNodes:
nano "$HADOOP_HOME/etc/hadoop/workers"

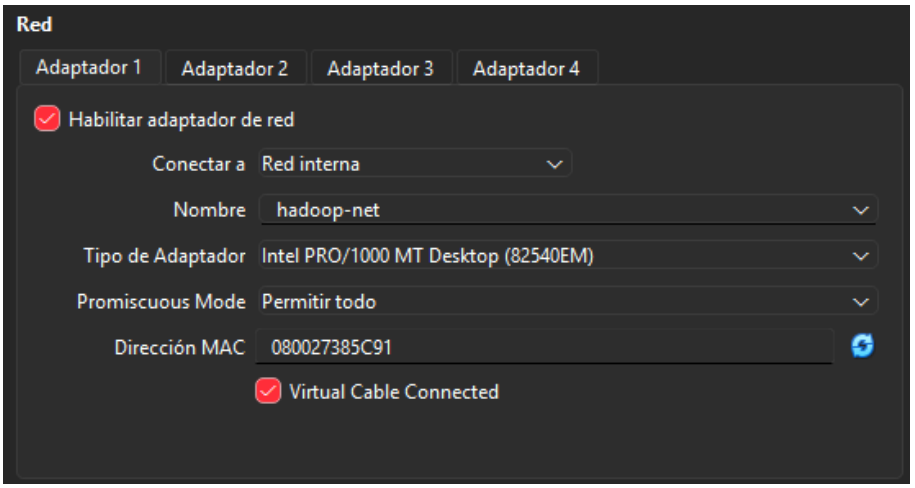
borramos localhost y ponemos:
datanode1
datanode2

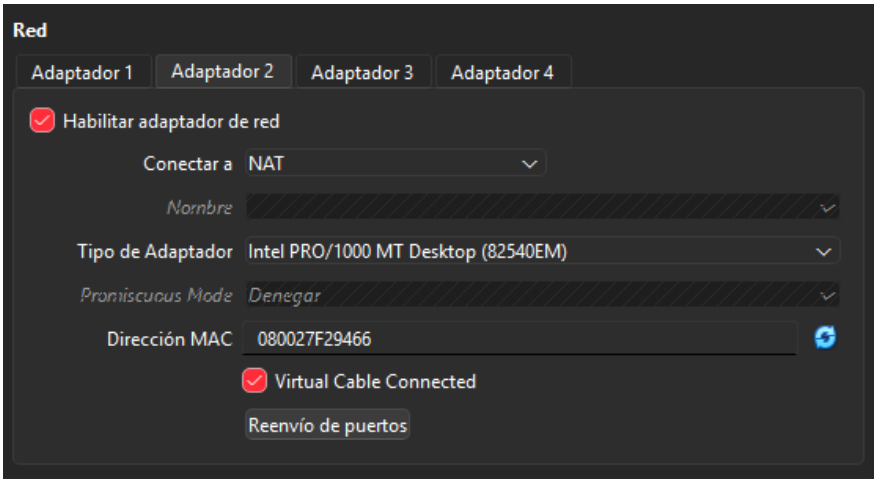
```
  GNU nano 8.7.1                /home/elias/hadoop-3.4.3/etc/hadoop/workers *
datanode1
datanode2
```

4. IP Fijas



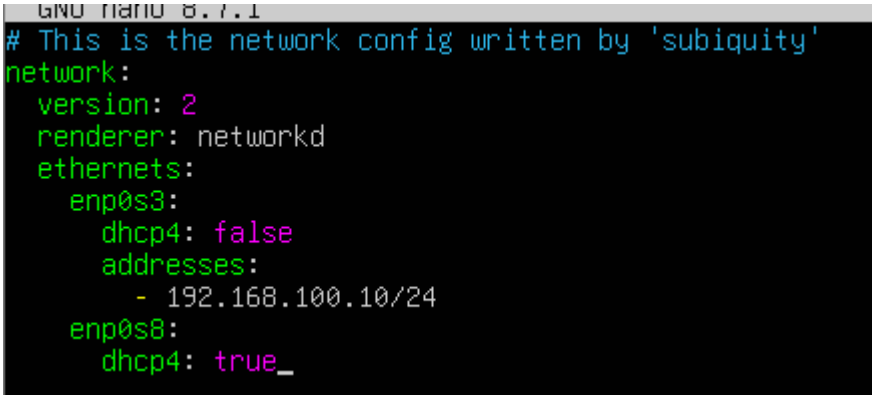
Para las máquinas master, datanode1 y 2 les ponemos estás configuraciones de adaptador:

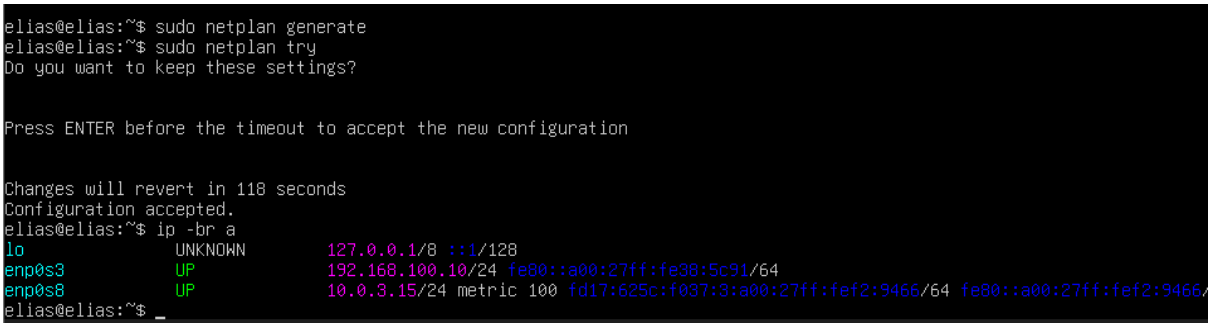


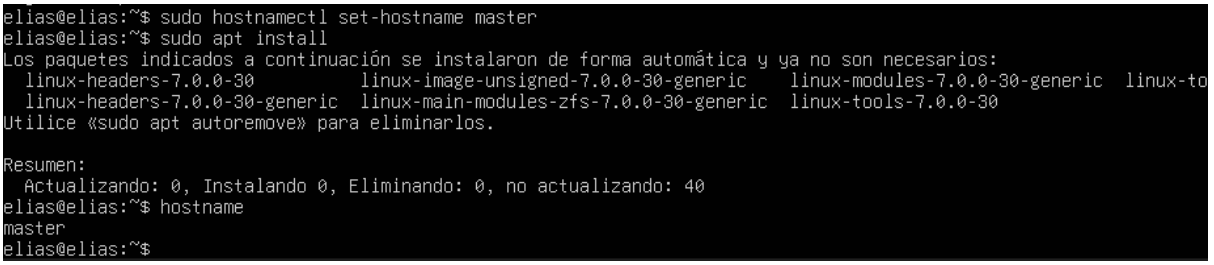


Nombre de la máquina	Dirección IP fija	Máscara de red
master	192.168.100.10	255.255.255.0
datanode1	192.168.100.11	255.255.255.0
datanode2	192.168.100.12	255.255.255.0

Para editar el fichero de netplan el comando es:
sudo nano /etc/netplan/00-installer-config.yaml
Master:







Datanode1:

```
# This is the network config written by 'subiquity'
network:
  version: 2
  renderer: networkd
  ethernets:
    enp0s3:
      dhcp4: false
      addresses:
        - 192.168.100.11/24
    enp0s8:
      dhcp4: true
```

```
elias@elias:~$ sudo netplan generate
elias@elias:~$ sudo netplan try
Do you want to keep these settings?


Press ENTER before the timeout to accept the new configuration


Changes will revert in 119 seconds
Configuration accepted.
elias@elias:~$ ip -br a
lo               UNKNOWN        127.0.0.1/8 ::1/128
enp0s3           UP             192.168.100.11/24 fe80::a00:27ff:fe3d:34af/64
enp0s8           UP             10.0.3.15/24 metric 100 fd17:625c:f037:3:a00:27ff:febb:7b85/64 fe80::a00:27ff:febb:7b85/
elias@elias:~$
```

```
elias@elias:~$ sudo hostnamectl set-hostname datanode1
elias@elias:~$ sudo apt install
Los paquetes indicados a continuación se instalaron de forma automática y ya no son necesarios:
  linux-headers-7.0.0-30          linux-image-unsigned-7.0.0-30-generic    linux-modules-7.0.0-30-generic  linux-tools
  linux-headers-7.0.0-30-generic  linux-main-modules-zfs-7.0.0-30-generic  linux-tools-7.0.0-30
Utilice «sudo apt autoremove» para eliminarlos.

Resumen:
  Actualizando: 0, Instalando 0, Eliminando: 0, no actualizando: 40
elias@elias:~$ hostname
datanode1
```

Datanode2:

```
# This is the network config written by 'subiquity'
network:
  version: 2
  renderer: networkd
  ethernets:
    enp0s3:
      dhcp4: false
      addresses:
        - 192.168.100.12/24
    enp0s8:
      dhcp4: true
```

```
elias@elias:~$ sudo netplan generate
elias@elias:~$ sudo netplan try
Do you want to keep these settings?


Press ENTER before the timeout to accept the new configuration


Changes will revert in 120 seconds
Configuration accepted.
elias@elias:~$ ip -br a
lo               UNKNOWN        127.0.0.1/8 ::1/128
enp0s3           UP             192.168.100.12/24 fe80::a00:27ff:fe9f:da95/64
enp0s8           UP             10.0.3.15/24 metric 100 fd17:625c:f037:3:a00:27ff:fe23:793e/64 fe80::a00:27ff:fe23:793e/64
elias@elias:~$ _
```

```
elias@elias:~$ sudo hostnamectl set-hostname datanode2
elias@elias:~$ sudo apt install
Los paquetes indicados a continuación se instalaron de forma automática y ya no son necesarios:
  linux-headers-7.0.0-30          linux-image-unsigned-7.0.0-30-generic    linux-modules-7.0.0-30-generic  linux-tools-7.0.
  linux-headers-7.0.0-30-generic  linux-main-modules-zfs-7.0.0-30-generic  linux-tools-7.0.0-30
Utilice «sudo apt autoremove» para eliminarlos.

Resumen:
  Actualizando: 0, Instalando 0, Eliminando: 0, no actualizando: 40
elias@elias:~$ hostname
datanode2
```

Ahora en las tres máquinas hay que modificar el archivo de host con el comando:
sudo nano /etc/hosts
para poner esto:
192.168.100.10 master
192.168.100.11 datanode1
192.168.100.12 datanode2

Conserva las líneas de localhost y de IPv6. Si aparece master, datanode1 o datanode2 en una línea con 127.0.1.1, quita ese nombre de esa línea, para que se resuelva mediante su IP interna.

```
127.0.0.1 localhost
127.0.1.1 elias
192.168.100.10 master
192.168.100.11 datanode1
192.168.100.12 datanode2
# The following lines are desirable for IPv6 capable hosts
::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
```

desde master hacemos ping a los dos datanode para ver si hay conexión entre ellos

```
elias@master:~$ ping -c 4 datanode1
PING datanode1 (192.168.100.11) 56(84) bytes of data.
64 bytes from datanode1 (192.168.100.11): icmp_seq=1 ttl=64 time=2.26 ms
64 bytes from datanode1 (192.168.100.11): icmp_seq=2 ttl=64 time=1.04 ms
64 bytes from datanode1 (192.168.100.11): icmp_seq=3 ttl=64 time=1.94 ms
64 bytes from datanode1 (192.168.100.11): icmp_seq=4 ttl=64 time=3.38 ms

--- datanode1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3054ms
rtt min/avg/max/mdev = 1.041/2.153/3.378/0.835 ms
elias@master:~$ ping -c 4 datanode2
PING datanode2 (192.168.100.12) 56(84) bytes of data.
64 bytes from datanode2 (192.168.100.12): icmp_seq=1 ttl=64 time=2.00 ms
64 bytes from datanode2 (192.168.100.12): icmp_seq=2 ttl=64 time=1.56 ms
64 bytes from datanode2 (192.168.100.12): icmp_seq=3 ttl=64 time=2.53 ms
64 bytes from datanode2 (192.168.100.12): icmp_seq=4 ttl=64 time=1.66 ms

--- datanode2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 5127ms
rtt min/avg/max/mdev = 1.562/1.936/2.534/0.380 ms
```

5. Regeneración de claves de servidor SSH en las máquinas clonadas
   Huella de la clave pública

```
elias@master:~$ ssh-keygen -lf /etc/ssh/ssh_host_ed25519_key.pub
256 SHA256:dSxBaLqD/vzwsIAqfjCt1NJGNvAthqahFcHri3hX/Yo root@elias (ED25519)
```

```
elias@datanode1:~$ ssh-keygen -lf /etc/ssh/ssh_host_ed25519_key.pub
256 SHA256:dSxBaLqD/vzwsIAqfjCt1NJGNvAthqahFcHri3hX/Yo root@elias (ED25519)
elias@datanode1:~$ _
```

```
elias@datanode2:~$ ssh-keygen -lf /etc/ssh/ssh_host_ed25519_key.pub
256 SHA256:dSxBaLqD/vzwsIAqfjCt1NJGNvAthqahFcHri3hX/Yo root@elias (ED25519)
elias@datanode2:~$ _
```

Vamos a eliminar las claves heredadas (como la hemos duplicado, ha cogido la de hadoop-base)

```
elias@master:~$ sudo rm /etc/ssh/ssh_host_*key*
elias@master:~$ sudo ssh-keygen -A
ssh-keygen: generating new host keys: RSA ECDSA ED25519
elias@master:~$ sudo systemctl restart ssh
elias@master:~$ ssh-keygen -lf /etc/ssh/ssh_host_ed25519_key.pub
256 SHA256:0WdnlXpp01fPTyHpKK4DBia9D7bsNy6Lkxq2tBXmvvc root@master (ED25519)
elias@master:~$ _
```

```
elias@datanode1:~$ ssh-keygen -lf /etc/ssh/ssh_host_ed25519_key.pub
256 SHA256:dSxBaLqD/vzwsIAqfjCt1NJGNvAthqahFcHri3hX/Yo root@elias (ED25519)
elias@datanode1:~$ sudo rm /etc/ssh/ssh_host_*key*
[sudo: authenticate] Password:
elias@datanode1:~$ sudo ssh-keygen -A
ssh-keygen: generating new host keys: RSA ECDSA ED25519
elias@datanode1:~$ ssh-keygen -lf /etc/ssh/ssh_host_ed25519_key.pub
256 SHA256:+F5raEv3fqoJeSVT00+HgcNv3AqNaI7WwszLQ9+Jc2A root@datanode1 (ED25519)
elias@datanode1:~$
```

```
elias@datanode2:~$ ssh-keygen -lf /etc/ssh/ssh_host_ed25519_key.pub
256 SHA256:dSxBaLqD/vzwsIAqfjCt1NJGNvAthqahFcHri3hX/Yo root@elias (ED25519)
elias@datanode2:~$ sudo rm /etc/ssh/ssh_host_*key*
elias@datanode2:~$ sudo ssh_keygen -A
sudo: 'ssh_keygen': command not found
elias@datanode2:~$ sudo ssh-keygen -A
ssh-keygen: generating new host keys: RSA ECDSA ED25519
elias@datanode2:~$ ssh-keygen -lf /etc/ssh/ssh_host_ed25519_key.pub
256 SHA256:cZ3OzU8sOp9948J7al32WxDYxqZLmAeiVq8zjDDgAS0 root@datanode2 (ED25519)
elias@datanode2:~$
```

Creamos una clave ssh y la copiamos en las demas:

```
elias@master:~$ ssh-keygen -t ed25519
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/elias/.ssh/id_ed25519):
Enter passphrase for "/home/elias/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Passphrases do not match.  Try again.
Enter passphrase for "/home/elias/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/elias/.ssh/id_ed25519
Your public key has been saved in /home/elias/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:/zP+AryuqNOVgBb4APmDdBLXqawBW0Pqe5Zkc/rhXjg elias@master
The key's randomart image is:
+--[ED25519 256]--+
|.=oo. .          |
|o+*..o           |
|+=++.o           |
|+.oo+ .          |
| .o* . .So       |
| .+ = . o.o      |
| . = E o  .o     |
|  o + =.  ..+    |
|    o*. .o..o=.  |
+----[SHA256]-----+
elias@master:~$
```

y la copiamos para datanode1 y 2

```
elias@master:~$ ssh-copy-id elias@master
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/elias/.ssh/id_ed25519.pub"
The authenticity of host 'master (192.168.100.10)' can't be established.
ED25519 key fingerprint is: SHA256:0WdnlXpp01fPTyHpKK4DBia9D7bsNy6Lkxq2tBXmvvc
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
elias@master's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'elias@master'"
and check to make sure that only the key(s) you wanted were added.
```

```
elias@master:~$ ssh-copy-id elias@datanode1
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/elias/.ssh/id_ed25519.pub"
The authenticity of host 'datanode1 (192.168.100.11)' can't be established.
ED25519 key fingerprint is: SHA256:+F5raEv3fqoJeSVT00+HgcNv3AqNaI7WwszLQ9+Jc2A
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
elias@datanode1's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'elias@datanode1'"
and check to make sure that only the key(s) you wanted were added.

elias@master:~$ ssh-copy-id elias@datanode2
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/elias/.ssh/id_ed25519.pub"
The authenticity of host 'datanode2 (192.168.100.12)' can't be established.
ED25519 key fingerprint is: SHA256:cZ3OzU8sOp9948J7al32WxDYxqZLmAeiVq8zjDDgAS0
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
elias@datanode2's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'elias@datanode2'"
and check to make sure that only the key(s) you wanted were added.

elias@master:~$
```

Comprobamos el acceso (para ello he tenido queeliminar la contraseña)

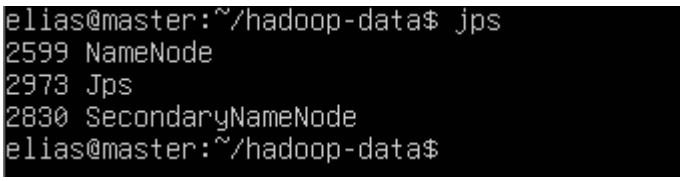
```
elias@master:~$ ssh-keygen -p -f "$HOME/.ssh/id_ed25519"
Enter old passphrase:
Key has comment 'elias@master'
Enter new passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved with the new passphrase.
```

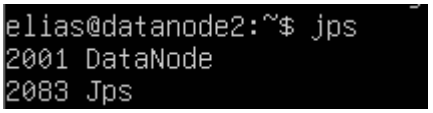
```
elias@master:~$ ssh -o BatchMode=yes elias@master hostname
master
elias@master:~$ ssh -o BatchMode=yes elias@datanode1 hostname
datanode1
elias@master:~$ ssh -o BatchMode=yes elias@datanode2 hostname
datanode2
elias@master:~$
```

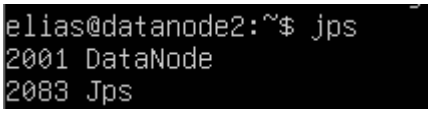
6. Inicialización y arranque de HDFS
   Inicializamos los metadatos del NameNode y crea la identidad del nuevo sistema HDFS

```
/************************************************************
SHUTDOWN_MSG: Shutting down NameNode at master/192.168.100.10
************************************************************/
elias@master:~$ ls
ab1234  ab1234.pub  hadoop-3.4.3  hadoop-3.4.3.tar.gz  hadoop-data
elias@master:~$ cd hadoop-data
elias@master:~/hadoop-data$ cd namenode/
elias@master:~/hadoop-data/namenode$ cd current
elias@master:~/hadoop-data/namenode/current$ ls
VERSION  fsimage_0000000000000000000  fsimage_0000000000000000000.md5  seen_txid
elias@master:~/hadoop-data/namenode/current$
```

Comprobamos que está funcionando:







Ahora desde master mostraremos el estado del HDFS

```
-------------------------------------------------
Live datanodes (2):

Name: 192.168.100.11:9866 (datanode1)
Hostname: datanode1
Decommission Status : Normal
Configured Capacity: 12040970240 (11.21 GB)
DFS Used: 24576 (24 KB)
Non DFS Used: 5712527360 (5.32 GB)
DFS Remaining: 5694451712 (5.30 GB)
DFS Used%: 0.00%
DFS Remaining%: 47.29%
Configured Cache Capacity: 0 (0 B)
Cache Used: 0 (0 B)
Cache Remaining: 0 (0 B)
Cache Used%: 100.00%
Cache Remaining%: 0.00%
Xceivers: 0
Last contact: Tue Nov 03 12:27:40 UTC 2026
Last Block Report: Tue Nov 03 12:21:27 UTC 2026
Num of Blocks: 0


Name: 192.168.100.12:9866 (datanode2)
Hostname: datanode2
Decommission Status : Normal
Configured Capacity: 12040970240 (11.21 GB)
DFS Used: 24576 (24 KB)
Non DFS Used: 5729320960 (5.34 GB)
DFS Remaining: 5677658112 (5.29 GB)
DFS Used%: 0.00%
DFS Remaining%: 47.15%
Configured Cache Capacity: 0 (0 B)
Cache Used: 0 (0 B)
Cache Remaining: 0 (0 B)
Cache Used%: 100.00%
Cache Remaining%: 0.00%
Xceivers: 0
Last contact: Tue Nov 03 12:27:39 UTC 2026
Last Block Report: Tue Nov 03 12:21:31 UTC 2026
Num of Blocks: 0
```

Prueba de escritura (desde master)
Ejecutamos el comando printf 'Prueba de funcionamiento de HDFS\n' > "$HOME/prueba-hdfs.txt"

```
elias@master:~/hadoop-data$ hdfs dfs -mkdir -p /practica
elias@master:~/hadoop-data$ hdfs dfs -put "$HOME/prueba-hdfs.txt" /practica
elias@master:~/hadoop-data$ hdfs dfs -ls /practica
Found 1 items
-rw-r--r--   2 elias supergroup         33 2026-11-03 12:31 /practica/prueba-hdfs.txt
```

```
elias@master:~/hadoop-data$ hdfs dfs -cat /practica/prueba-hdfs.txt
Prueba de funcionamiento de HDFS
```

Comprobamos el estado del archivo:

```
elias@master:~/hadoop-data$ hdfs fsck /practica/prueba-hdfs.txt -files -blocks -locations
Connecting to namenode via http://master:9870/fsck?ugi=elias&files=1&blocks=1&locations=1&path=%2Fpractica%2Fprueba-hdfs
FSCK started by elias (auth:SIMPLE) from /192.168.100.10 for path /practica/prueba-hdfs.txt at Tue Nov 03 12:35:04 UTC 2

/practica/prueba-hdfs.txt 33 bytes, replicated: replication=2, 1 block(s):  OK
0. BP-1231486192-192.168.100.10-1793708212407:blk_1073741825_1001 len=33 Live_repl=2  [DatanodeInfoWithStorage[192.168.1
8977f786fa,DISK], DatanodeInfoWithStorage[192.168.100.11:9866,DS-a5422e07-4923-4f4f-9cc8-e4f5fc7cedfc,DISK]]


Status: HEALTHY
 Number of data-nodes:  2
 Number of racks:               1
 Total dirs:                    0
 Total symlinks:                0

Replicated Blocks:
 Total size:    33 B
 Total files:   1
 Total blocks (validated):      1 (avg. block size 33 B)
 Minimally replicated blocks:   1 (100.0 %)
 Over-replicated blocks:        0 (0.0 %)
 Under-replicated blocks:       0 (0.0 %)
 Mis-replicated blocks:         0 (0.0 %)
 Default replication factor:    2
 Average block replication:     2.0
 Missing blocks:                0
 Corrupt blocks:                0
 Missing replicas:              0 (0.0 %)
 Blocks queued for replication: 0

Erasure Coded Block Groups:
 Total size:    0 B
 Total files:   0
 Total block groups (validated):        0
 Minimally erasure-coded block groups:  0
 Over-erasure-coded block groups:       0
 Under-erasure-coded block groups:      0
 Unsatisfactory placement block groups: 0
 Average block group size:      0.0
 Missing block groups:          0
 Corrupt block groups:          0
 Missing internal blocks:       0
 Blocks queued for replication: 0
FSCK ended at Tue Nov 03 12:35:04 UTC 2026 in 77 milliseconds

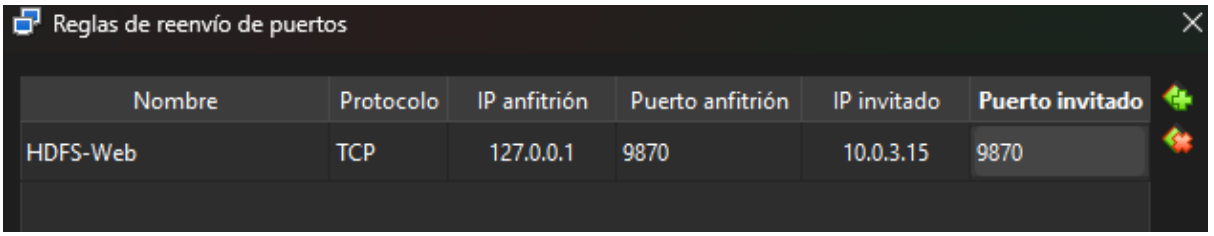

The filesystem under path '/practica/prueba-hdfs.txt' is HEALTHY
elias@master:~/hadoop-data$ _
```

Comprobamos la descarga del archivo:

```
elias@master:~/hadoop-data$ hdfs dfs -get /practica/prueba-hdfs.txt "$HOME/prueba-recuperada.txt"
elias@master:~/hadoop-data$ _
```

```
elias@master:~/hadoop-data$ diff "$HOME/prueba-hdfs.txt" "$HOME/prueba-recuperada.txt"
elias@master:~/hadoop-data$ _
```

7. Configuración del acceso web a HDFS mediante reenvío de puertos
   Agregamos una nueva regla al adaptador LAN (el 2)



Overview 'master:9000' (✔active)

Started:	Tue Nov 03 13:21:05 +0100 2026
Version:	3.4.3, r9d50c6884666e794e45102260a4017bb31802e1b
Compiled:	Fri Feb 13 15:23:00 +0100 2026 by stevel from branch-3.4.3
Cluster ID:	CID-f3dd99f3-2e97-4cfb-80d6-58d4fc217592
Block Pool ID:	BP-1231486192-192.168.100.10-1793708212407

Summary

Security is off.

Safemode is off.

3 files and directories, 1 blocks (1 replicated blocks, 0 erasure coded block groups) = 4 total filesystem object(s).

Heap Memory used 92.89 MB of 291 MB Heap Memory. Max Heap Memory is 1.81 GB.

Non Heap Memory used 68.26 MB of 71.05 MB Commited Non Heap Memory. Max Non Heap Memory is <unbounded>.

Configured Capacity:	22.43 GB
Configured Remote Capacity:	0 B
DFS Used:	88 KB (0%)
Non DFS Used:	10.66 GB
DFS Remaining:	10.59 GB (47.22%)
Block Pool Used:	88 KB (0%)
DataNodes usages% (Min/Median/Max/stdDev):	0.00% / 0.00% / 0.00% / 0.00%
Live Nodes	2 (Decommissioned: 0, In Maintenance: 0)
Dead Nodes	0 (Decommissioned: 0, In Maintenance: 0)
Decommissioning Nodes	0
Entering Maintenance Nodes	0
Total Datanode Volume Failures	0 (0 B)
Number of Under-Replicated Blocks	0
Number of Blocks Pending Deletion (including replicas)	0
Block Deletion Start Time	Tue Nov 03 13:21:05 +0100 2026
Last Checkpoint Time	Tue Nov 03 13:16:52 +0100 2026
Last HA Transition Time	Never
Enabled Erasure Coding Policies	RS-6-3-1024k

NameNode Journal Status

Current transaction ID: 10

Journal Manager	State
FileJournalManager(root=/home/elias/hadoop-data/namenode)	EditLogFileOutputStream(/home/elias/hadoop-data/namenode/current/edits_inprogress_0000000000000000010)

NameNode Storage

Storage Directory	Type	State
/home/elias/hadoop-data/namenode	IMAGE_AND_EDITS	Active

DFS Storage Types

Storage Type	Configured Capacity	Capacity Used	Capacity Remaining	Block Pool Used	Nodes In Service
DISK	22.43 GB	88 KB (0%)	10.59 GB (47.22%)	88 KB	2

Hadoop, 2026.